

Problem Set 3

Applied Stats/Quant Methods 1

Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in **R** using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

Table 1: Regression Results for Model 1: voteshare ~ difflog

	<i>Dependent variable:</i>
	Incumbent Vote Share
Spending Difference (difflog)	0.0417*** (0.0010)
Constant	0.5790*** (0.0023)
Observations	3,193
R ²	0.3673
Adjusted R ²	0.3671
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

```

1 #Question1.1
2 inc.sub <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/StatsI_
   2025/main/datasets/incumbents_subset.csv")
3 head(inc.sub)
4 summary(inc.sub)
5 model <- lm(voteshare ~ difflog, data = inc.sub)
6 summary(model)
7 library(stargazer)
8 stargazer(model,
9           type = "latex",
10          title = "Regression Results for Model 1: voteshare ~ difflog",
11          dep.var.labels = "Incumbent Vote Share",
12          covariate.labels = c("Spending Difference (difflog)"),
13          digits = 4,
14          omit.stat = c("f", "ser"),
15          align = TRUE,
16          no.space = TRUE)

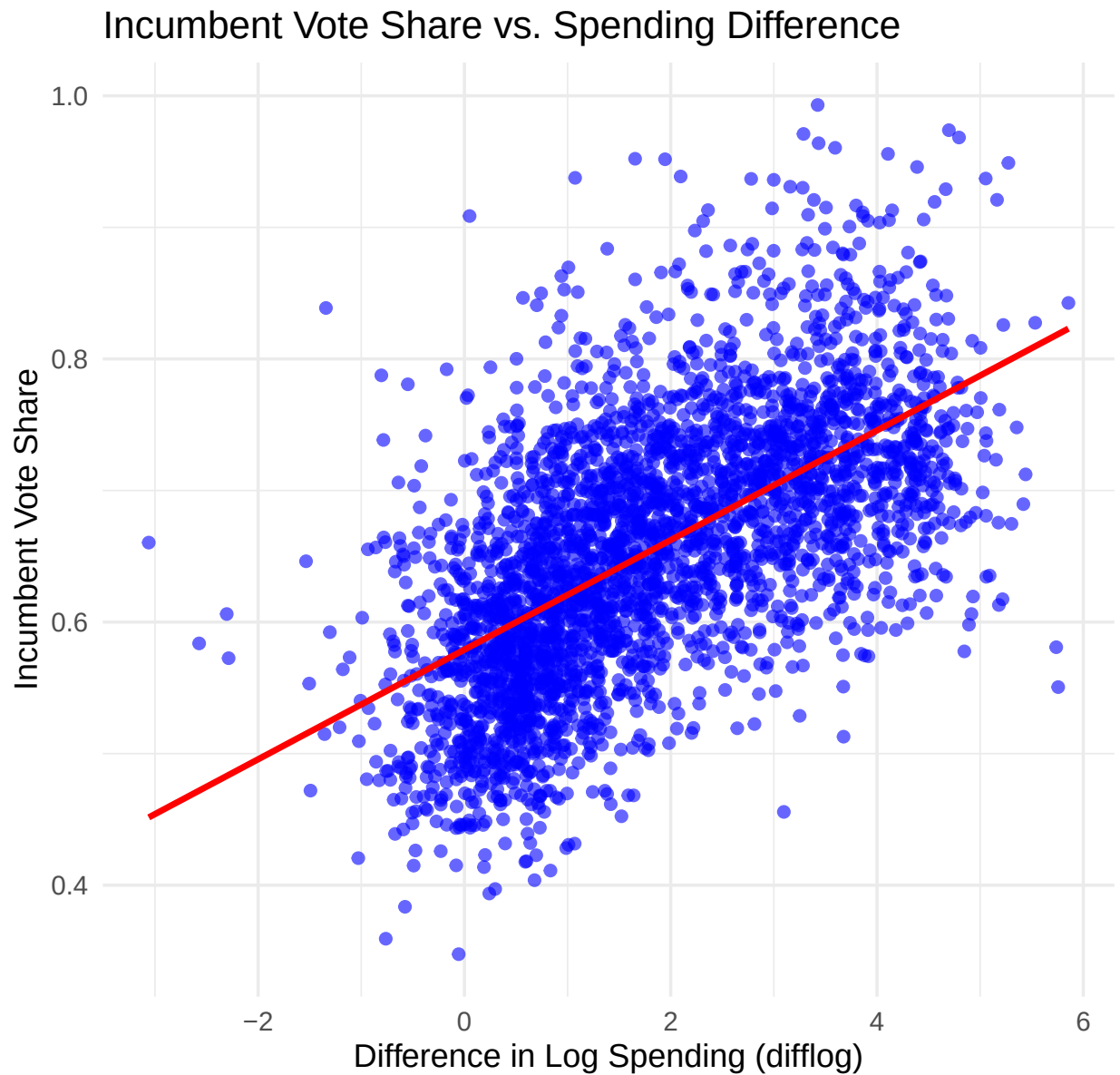
```

2. Make a scatterplot of the two variables and add the regression line.

```

1 #Question1.2
2 library(ggplot2)
3 ggplot(inc.sub, aes(x = difflog, y = voteshare)) +
4   geom_point(color = "blue", alpha = 0.6) +
5   geom_smooth(method = "lm", se = FALSE, color = "red") +
6   labs(
7     title = "Incumbent Vote Share vs. Spending Difference",
8     x = "Difference in Log Spending (difflog)",
9     y = "Incumbent Vote Share"
10  ) +
11  theme_minimal()

```



3. Save the residuals of the model in a separate object.

```
1 #Question1.3  
2 residuals_model <- resid(model)  
3 head(residuals_model)
```

4. Write the prediction equation.

```
1 #Question1.4  
2 coef(model)
```

$$\widehat{voteshare} = 0.58 + 0.04 \times \text{difflog}$$

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

```
1 #Question2.1
2 model2 <- lm(presvote ~ difflog, data = inc.sub)
3 summary(model2)
4 stargazer(model2,
5           type = "latex",
6           title = "Regression Results for Model 2: presvote ~ difflog",
7           dep.var.labels = "presvote",
8           covariate.labels = c("Spending Difference (difflog)"),
9           digits = 4,
10          omit.stat = c("f", "ser"),
11          align = TRUE,
12          no.space = TRUE)
```

Table 2: Regression Results for Model 2: presvote difflog

	<i>Dependent variable:</i>
	presvote
Spending Difference (difflog)	0.0238*** (0.0014)
Constant	0.5076*** (0.0032)
Observations	3,193
R ²	0.0880
Adjusted R ²	0.0877

Note: *p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

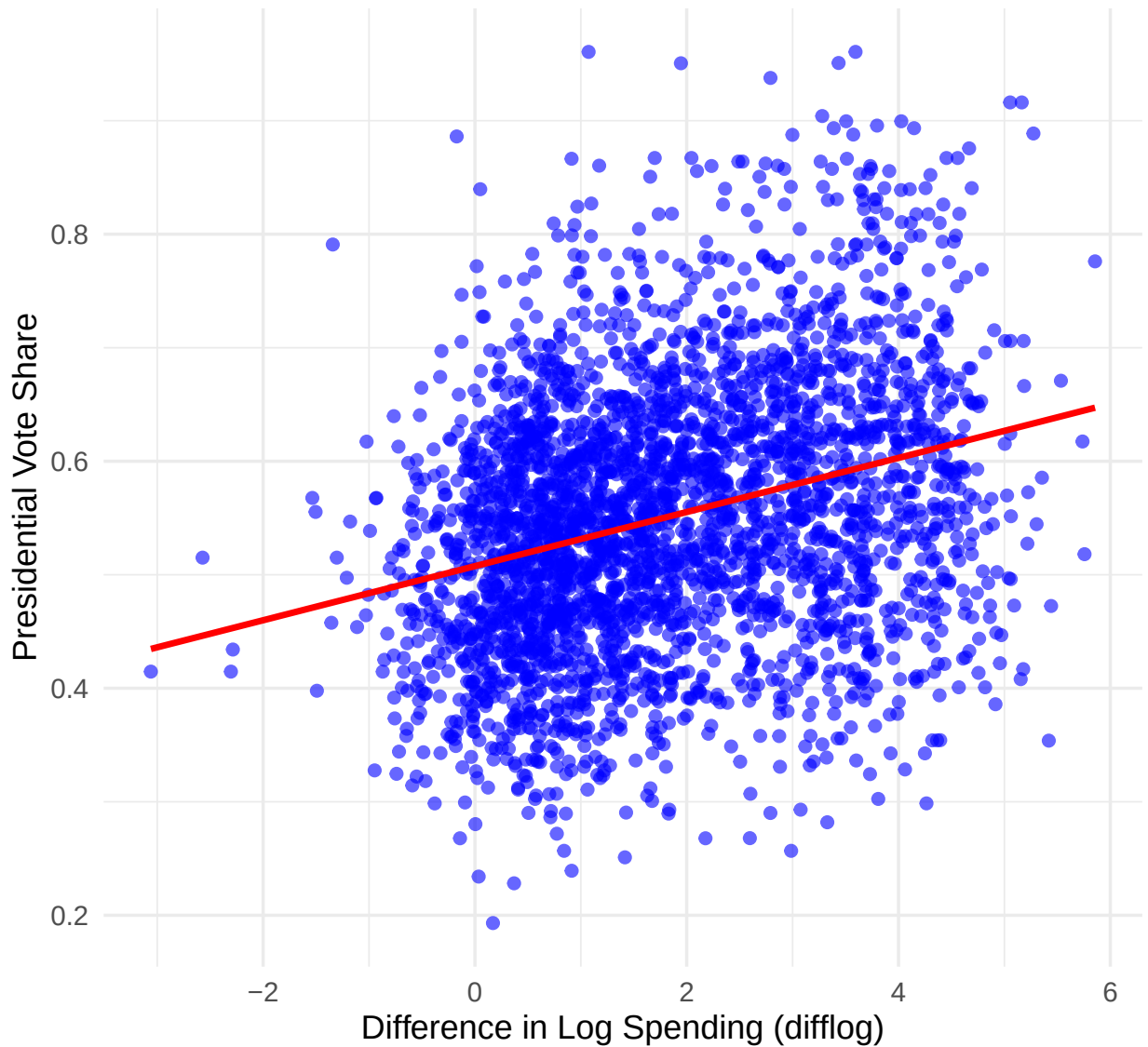
```
1 #Question2.2
2 ggplot(inc.sub, aes(x = difflog, y = presvote)) +
3   geom_point(color = "blue", alpha = 0.6) +
4   geom_smooth(method = "lm", se = FALSE, color = "red") +
5   labs(
6     title = "Presidential Vote Share vs. Spending Difference",
```

```

7   x = "Difference in Log Spending (difflog)",
8   y = "Presidential Vote Share"
9 ) +
10 theme_minimal()

```

Presidential Vote Share vs. Spending Difference



3. Save the residuals of the model in a separate object.

```

1 #Question2.3
2 residuals_model2 <- resid(model2)
3 head(residuals_model2)

```

4. Write the prediction equation.

```
1 #Question2.4
2 coef(model2)
```

$$\widehat{presvote.} = 0.51 + 0.02 \times \text{difflog}$$

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

```
1 #Question3.1
2 model3 <- lm(voteshare ~ presvote, data = inc.sub)
3 summary(model3)
4 stargazer(model3,
5           type = "latex",
6           title = "Regression Results for Model 3: voteshare ~ presvote",
7           dep.var.labels = "Incumbent Vote Share",
8           covariate.labels = c("presvote"),
9           digits = 4,
10          omit.stat = c("f", "ser"),
11          align = TRUE,
12          no.space = TRUE)
```

Table 3: Regression Results for Model 3: voteshare ~ presvote

<i>Dependent variable:</i>	
Incumbent Vote Share	
presvote	0.3880*** (0.0135)
Constant	0.4413*** (0.0076)
Observations	3,193
R ²	0.2058
Adjusted R ²	0.2056
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	

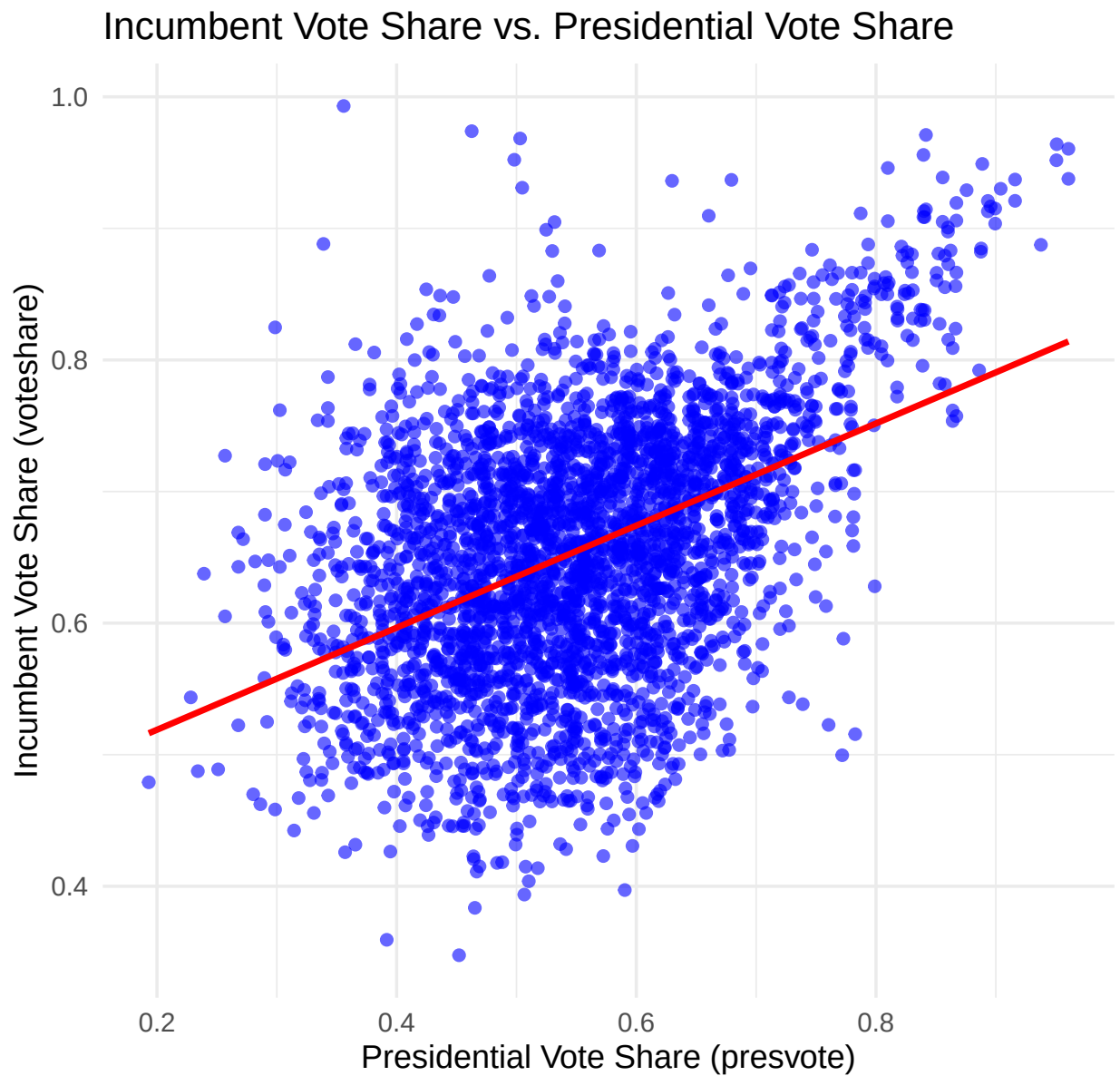
2. Make a scatterplot of the two variables and add the regression line.

```
1 #Question3.2
2 ggplot(inc.sub, aes(x = presvote, y = voteshare)) +
3   geom_point(color = "blue", alpha = 0.6) +
4   geom_smooth(method = "lm", se = FALSE, color = "red") +
5   labs(
6     title = "Incumbent Vote Share vs. Presidential Vote Share",
7     x = "Presidential Vote Share (presvote)",
```

```

8   y = "Incumbent Vote Share (voteshare)"
9   ) +
10  theme_minimal()

```



3. Write the prediction equation.

```

1 #Question3.3
2 coef(model3)

```

$$\widehat{voteshare} = 0.44 + 0.39 \times presvote$$

Question 4

The residuals from part (a) tell us how much of the variation in **voteshare** is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in **presvote** is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```

1 #Question4.1
2 residuals_model_reg <- lm(residuals_model ~ residuals_model2)
3 summary(residuals_model_reg)
4 stargazer(residuals_model_reg,
5           type = "latex",
6           title = "Regression Results for Model 4: residual_model ~
7           residuals_model2",
8           dep.var.labels = "residual_model",
9           covariate.labels = c("residual_model2"),
10          digits = 4,
11          omit.stat = c("f", "ser"),
12          align = TRUE,
13          no.space = TRUE)

```

Table 4: Regression Results for Model 4: residual model residuals model2

	Dependent variable:
	residual model
residual model2	0.2569*** (0.0118)
Constant	-0.0000 (0.0013)
Observations	3,193
R ²	0.1300
Adjusted R ²	0.1298
Note: *p<0.1; **p<0.05; ***p<0.01	

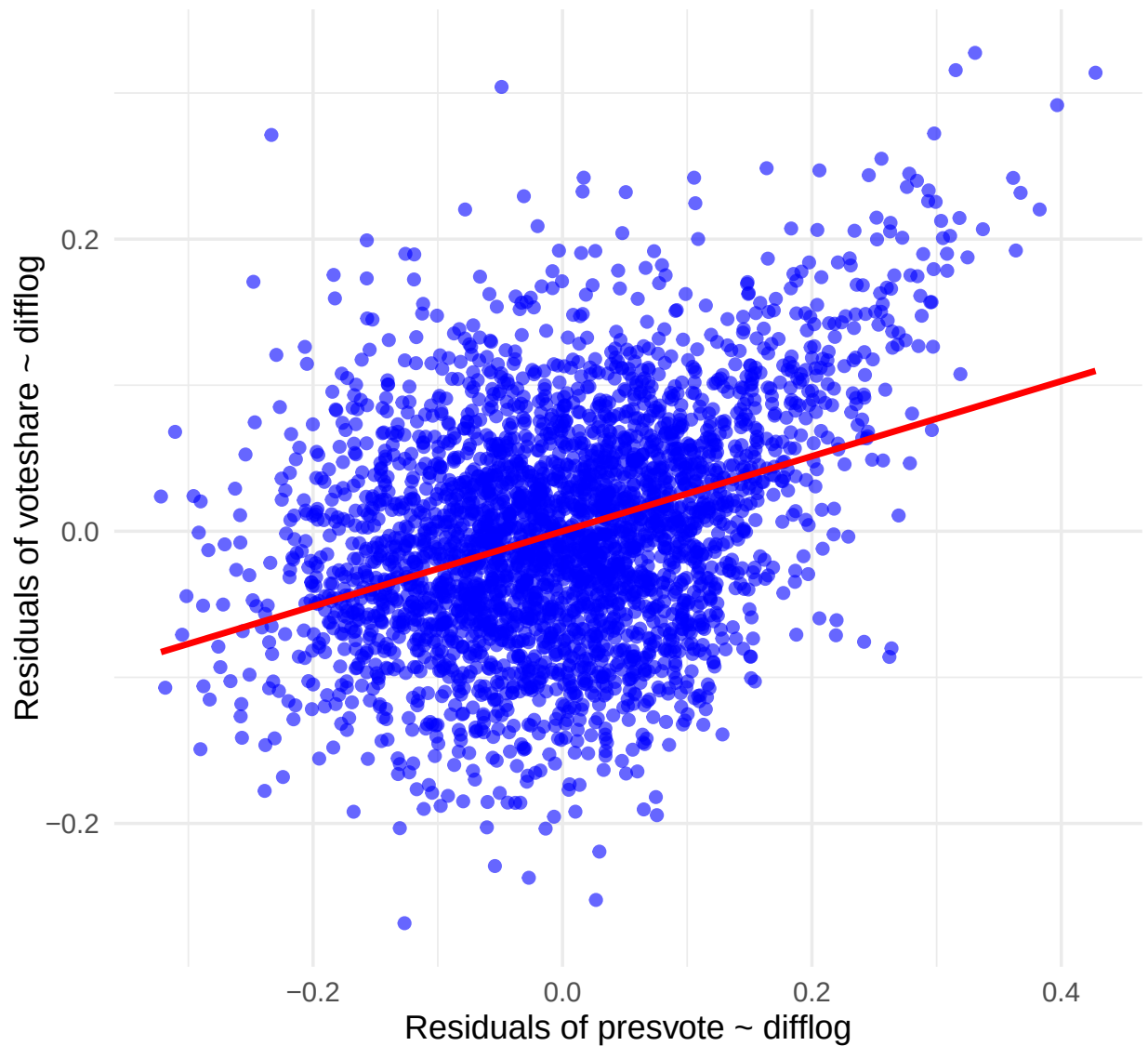
2. Make a scatterplot of the two residuals and add the regression line.

```

1 #Question4.2
2 ggplot(data = inc.sub, aes(x = residuals_model2, y = residuals_model)) +
3   geom_point(color = "blue", alpha = 0.6) +
4   geom_smooth(method = "lm", se = FALSE, color = "red") +
5   labs(
6     title = "Add-Variable Plot: voteshare residuals vs presvote residuals",
7     x = "Residuals of presvote ~ difflog",
8     y = "Residuals of voteshare ~ difflog"
9   ) +
10  theme_minimal()

```

Add-Variable Plot: voteshare residuals vs presvote residuals



3. Write the prediction equation.

```
1 #Question4.3
2 coef(residuals_model_reg)
3 round(coef(residuals_model_reg), 4)
```

$$\widehat{residualmodel} = 0 + 0.26 \times residual\ model2$$

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```

1 #Question5.1
2 model4 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
3 summary(model4)
4 stargazer(model4,
5           type = "latex",
6           title = "Regression Results for Model 4: voteshare ~ difflog +
7           presvote",
8           dep.var.labels = "Incumbent Vote Share",
9           covariate.labels = c("Spending Difference (difflog) + presvote"
10          ),
11          digits = 4,
12          omit.stat = c("f", "ser"),
13          align = TRUE,
14          no.space = TRUE)

```

Table 5: Regression Results for Model 4: `voteshare ~ difflog + presvote`

	<i>Dependent variable:</i>
	Incumbent Vote Share
Spending Difference (difflog) + presvote	0.0355*** (0.0009)
presvote	0.2569*** (0.0118)
Constant	0.4486*** (0.0063)
Observations	3,193
R ²	0.4496
Adjusted R ²	0.4493

Note:

*p<0.1; **p<0.05; ***p<0.01

2. Write the prediction equation.

```
1 #Question5.2
2 coef(model4)
```

$$\widehat{voteshare} = 0.49 + 0.036 \times \text{difflog} + 0.26 \times \text{presvote}$$

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

The identical value in question 4 and 5 is the coefficient of presvote (0.26). This is because the effect of presvote on voteshare, controlling for difflog, is the same whether you include difflog directly in a multiple regression or “remove” its influence first and then regress the residuals.